

5	(a)	(i)	Hypothetical <u>gas obeys equation of state $pV = nRT$</u> (perfectly at all pressures, temperature s and volume)	B1
		(ii)	Mean-square-speed (of atoms / molecules)	B1
		(iii)	$p = \frac{1}{3} \rho \langle c^2 \rangle$	
			$p = \frac{Nm}{V}$ with N explained (m = mass of a molecule) Or $p = \frac{M}{V}$ (M = mass of a gas)	B1
			$pV = \frac{1}{3} Nm \langle c^2 \rangle$	B1
			$pV = NkT$ with p, V, T explained	B1
			So mean kinetic energy $\langle E_k \rangle = \frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$	B1
	(b)	(i)	Internal energy U of a system is <u>sum of a random distribution of kinetic and potential energies associated with the molecules of a system.</u>	B1
		(ii)	(in ideal gas) no intermolecular forces, hence no potential energy	B1
			Internal energy is (solely) kinetic energy (of particles)	
			Since <u>mean (translational) kinetic energy is proportional to thermodynamic temperature of the gas</u> , the internal energy is directly proportional as well.	B1

(c)

	Increase in internal energy / J	Heat supplied to gas / J	Work done on gas / J
A to B	1200	0	1200
B to C	-1350	-1350	0
C to D	-600	0	-600
D to A	750	750	0

1 if W_{BC} and $W_{DA} = 0$

1 if ΔU_{DA} calculated correctly.

1 if W_{AB} and W_{CD} , Q_{BC} and Q_{DA} calculated correctly